

Restaurant Review Sentiment Analysis

Midterm Project Report

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Introduction and Motivation:

In today's digital world, customer reviews play a major role in shaping perceptions of restaurants and influencing customer decisions. Websites like Yelp, TripAdvisor, and Google Reviews have become essential platforms where people express their satisfaction or disappointment about dining experiences. However, manually reading and understanding thousands of reviews is impractical. Artificial Intelligence (AI) and Natural Language Processing (NLP) make it possible to automate this process by classifying reviews as positive, negative or neutral.

The motivation behind this project lies in the growing importance of sentiment analysis in real-world applications. By creating a system that automatically interprets customer feedback, we can provide restaurant owners, managers and even customers with insights into overall satisfaction trends and service quality.



Project Goals:

The main goal of this project is to develop a sentiment analysis model that accurately classifies restaurant reviews using AI and NLP techniques. The system should identify the underlying emotion behind each review and provide interpretable insights for decision-making. Another goal is to explore and compare how classical machine learning models perform versus modern transformer-based models such as BERT.

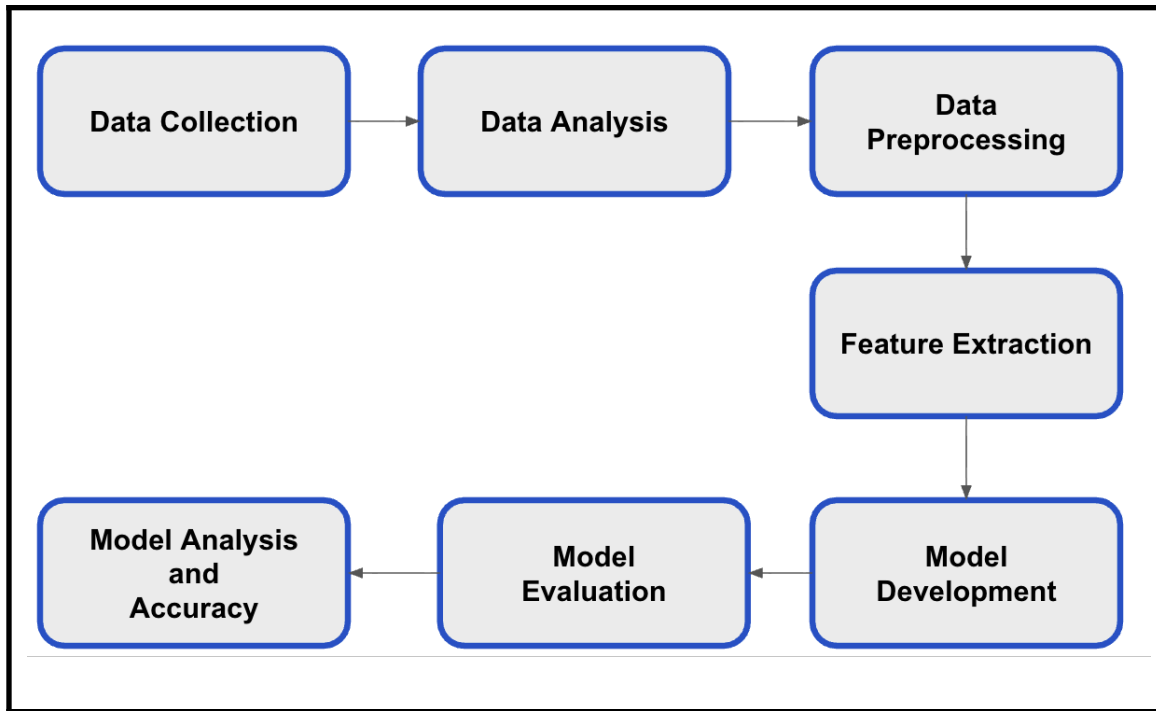
Objectives:

The key objectives of this project include:

- Collecting restaurant review data from reliable sources.
- Preprocessing the data by cleaning and structuring text for model input.
- Experimenting with different feature extraction techniques such as TF-IDF and word embeddings.
- Building and evaluating various AI models for sentiment classification.
- Measuring performance using metrics like accuracy, recall, precision, and F1-score.
- Identifying challenges in interpreting nuanced emotions such as sarcasm or mixed sentiments.

Basic Approach and Methodology:

The project follows a multi-step AI pipeline that includes data collection, exploration, preprocessing, model development, evaluation, and analysis. Each stage plays a vital role in creating a reliable and effective sentiment analysis system.



Data Collection

The dataset will consist of restaurant reviews collected from sources such as Yelp, TripAdvisor, and Kaggle. Each review will have associated sentiment labels for supervised learning. Data will be cleaned to remove duplicates and inconsistencies.

Exploratory Data Analysis (EDA)

EDA will help identify sentiment distributions, review length variations and frequent keywords. It provides an early understanding of the dataset and assists in identifying any biases or anomalies.

Data Preprocessing

Data preprocessing involves cleaning text by removing punctuation, numbers and unnecessary symbols. Tokenization, stop-word removal and lemmatization will be performed to standardize words. The goal is to prepare consistent input for feature extraction.

Feature Extraction and Representation

We will experiment with TF-IDF, Word2Vec and transformer-based embeddings like BERT to represent textual data numerically. These representations help capture semantic meaning and context.

Model Development

Multiple machine learning algorithms such as Naïve Bayes, Logistic Regression and SVM will be implemented as baselines. Deep learning models including LSTM and BERT will be tested for advanced contextual understanding. Hyperparameter optimization will ensure robust model performance.

Evaluation Metrics

Model performance will be assessed through Accuracy, Precision, Recall, F1-Score, and Confusion Matrix. Cross-validation will be used to verify the consistency of results.

Model Interpretability

Model explainability will be analyzed through feature importance and attention visualization for transformer-based architectures.

Evaluation Strategy and Validation Techniques:

To ensure model reliability, the dataset will be divided into training, validation, and testing sets. Stratified k-fold cross-validation will be used to ensure balanced class representation. We will analyze performance variations across folds and utilize ROC-AUC and confusion matrices to validate predictive power. The goal is to select the model that achieves the best balance between accuracy and generalization.

Anticipated Results:

We expect the sentiment analysis model to accurately classify restaurant reviews with high reliability. Transformer-based models such as BERT are anticipated to outperform traditional ML methods due to their superior ability to capture

contextual meaning. The final model should provide actionable insights about customer satisfaction and recurring feedback themes.

Real-Time Application Purpose:

The purpose of developing this sentiment analysis system is to enable real-time understanding of customer feedback. In practice, this technology can help restaurants monitor performance, detect negative reviews immediately, and identify improvement areas. Some real-world applications include:

- Enhancing customer experience by resolving issues faster.
- Monitoring brand reputation across review platforms.
- Optimizing menu items based on feedback trends.
- Comparing competitor reviews for market analysis.
- Integrating with recommendation systems to improve personalized suggestions.

By combining AI and NLP, the project demonstrates how data-driven solutions can transform customer engagement and business decision-making.

Ethical Considerations and Limitations:

Despite its usefulness, sentiment analysis has ethical and technical limitations. Models may inherit biases present in training data, leading to skewed interpretations. Moreover, AI systems may misinterpret sarcasm or cultural nuances. Ethical data sourcing, transparency in algorithmic decisions, and continued model monitoring are essential to ensure fairness and reliability.

Conclusion:

This project demonstrates the value of AI and NLP in automating sentiment analysis for restaurant reviews. By leveraging advanced modeling techniques, it bridges the gap between human language and machine interpretation. The outcome aims to showcase how AI can provide meaningful insights for customer satisfaction, brand management, and market competitiveness.

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